

Diode Wave Shaping Techniques, Clipping and Clamping Circuits, Voltage Regulation using Zener Diode

Text Books

1. Electronic Devices and Circuit Theory

by R Boylestad and L Nashelsky

2. Op-Amps and Linear Integrated Circuits

by Ramakant A. Gayakwad

3. Microelectronic Circuits Analysis and Design

by Muhammad H. Rashid

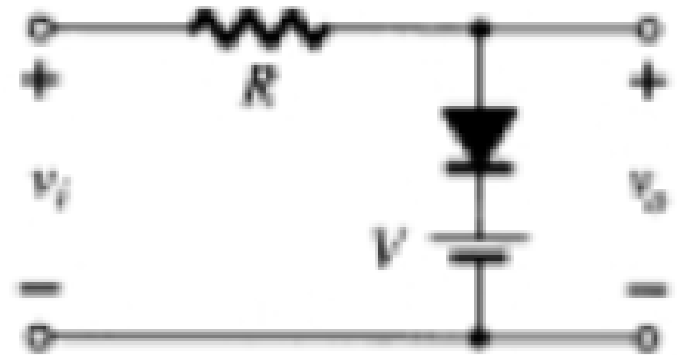
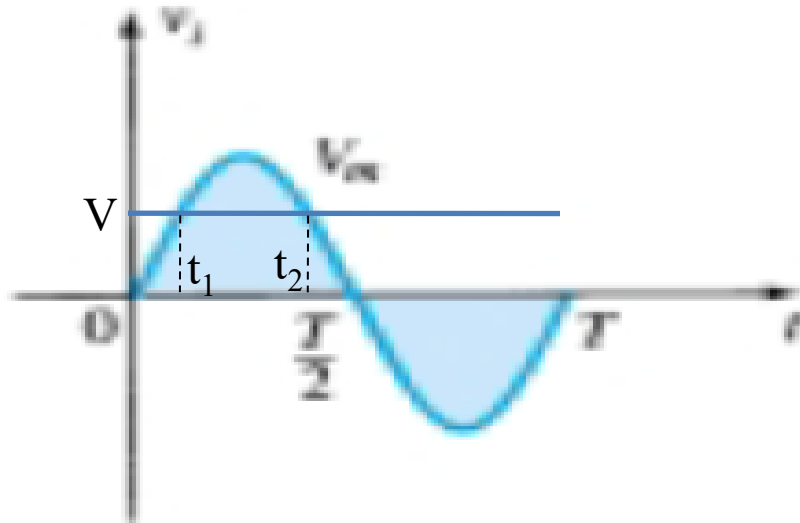
4. Electronic Principles 7th Edition

by Albert Malvino, David Bates

Clipper circuit

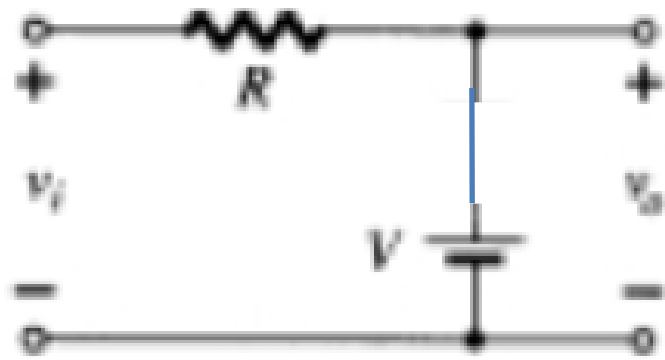
- Used to clip off a portion of the input signal.
- Needs diode, resistor and dc battery.
- Half wave rectifier is the simplest form of clipper which clips off the entire half cycle of the input.

Example-1

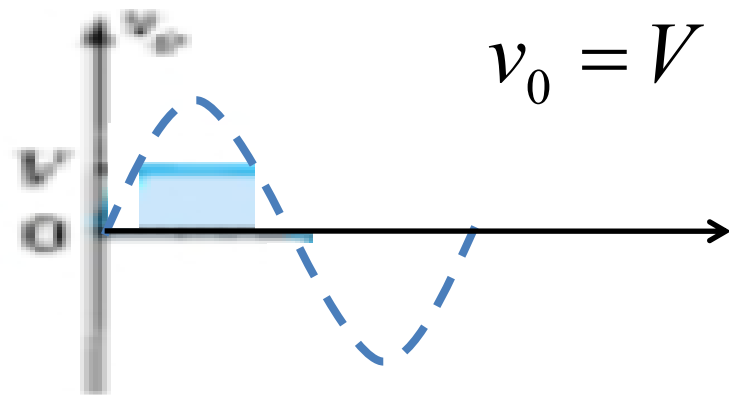
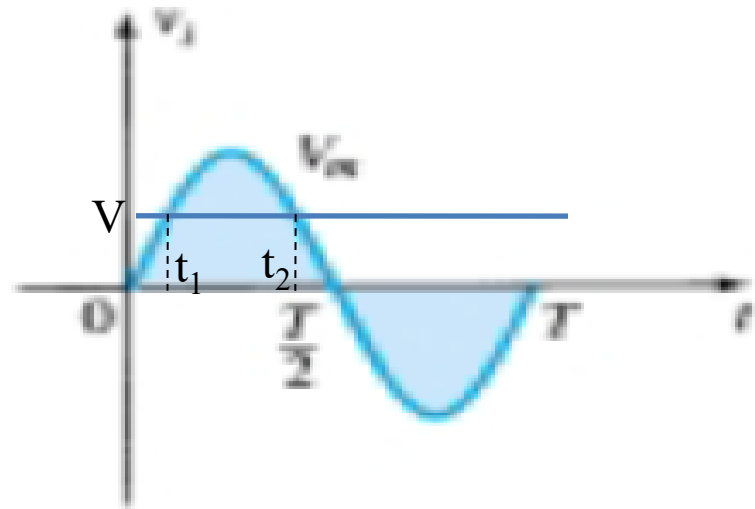


$$v_o = V$$

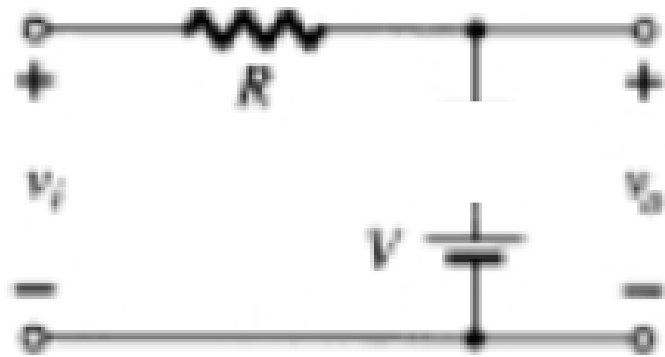
Example-1



The diode will turn on during the positive half cycle only when $v_i \geq V$, the output is then $v_o = V$.

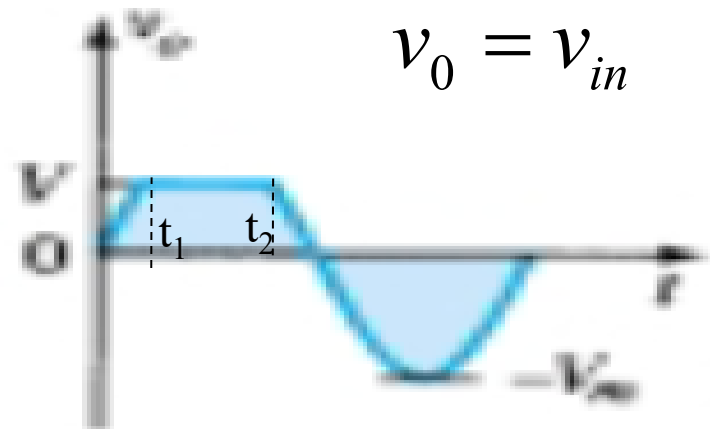
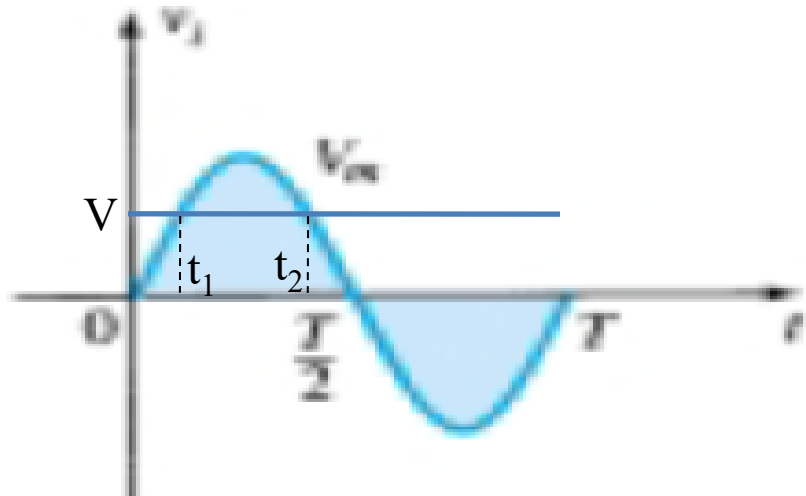


Example-1



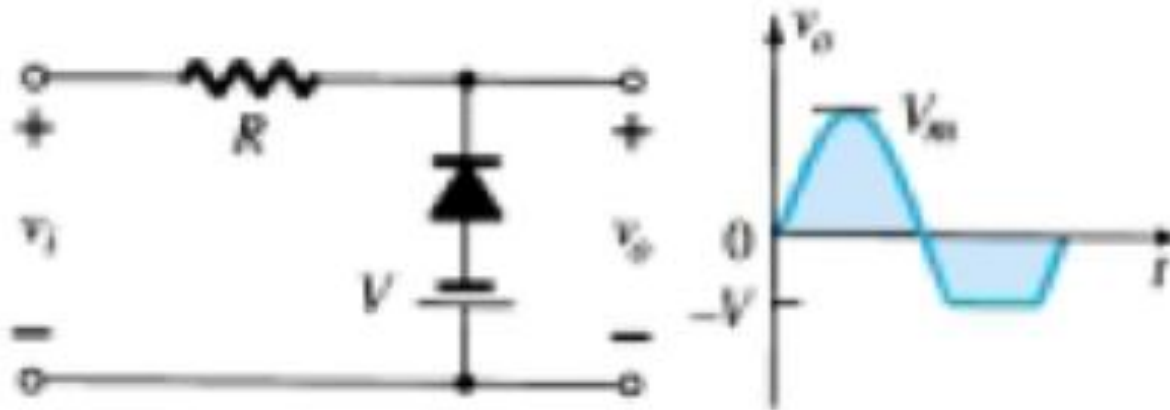
➤ For $v_i < V$, the diode is off, the diode will also be off during entire negative half cycle of the input signal.

➤ When the diode is off, $v_o = v_i$.



$$v_o = v_{in}$$

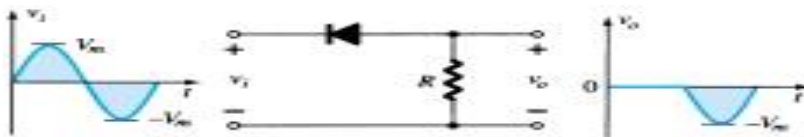
Example-2



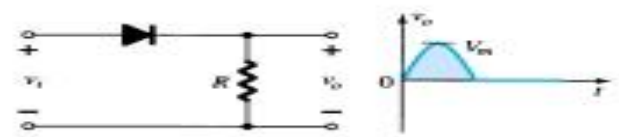
- The diode is off during entire positive half cycle of the input signal. Therefore, $v_o = v_i$.
- In negative half cycle the diode will be also off for $v_i < -V$, output will be also, $v_o = v_i$.
- The diode will turn on when $v_i \geq -V$, the output is then $v_o = -V$.

Simple Series Clippers (Ideal Diodes)

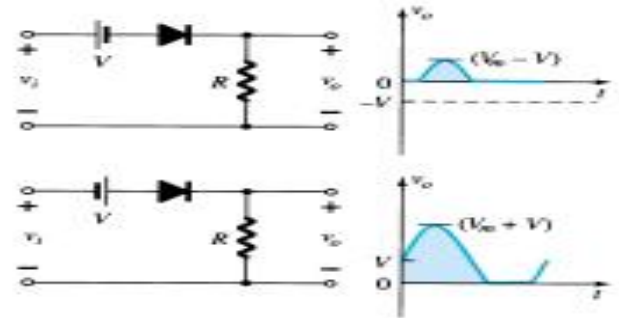
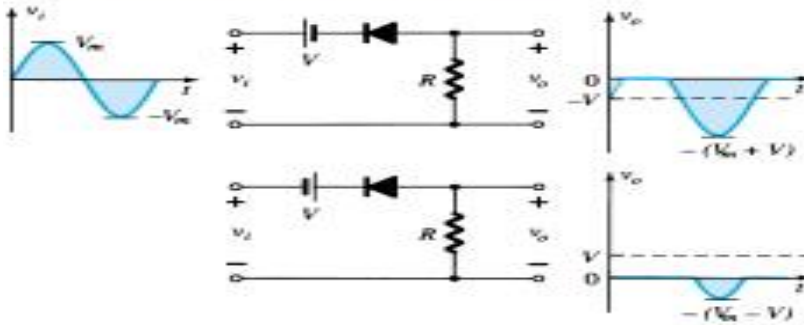
POSITIVE



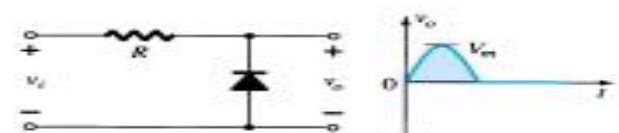
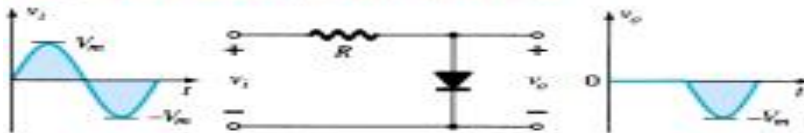
NEGATIVE



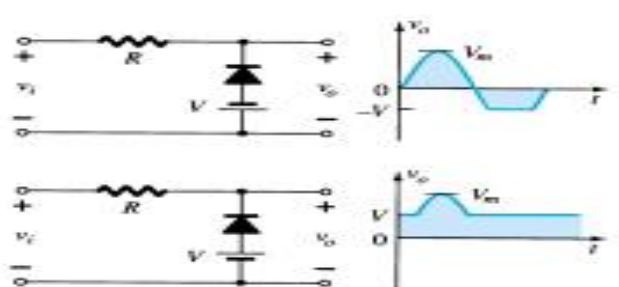
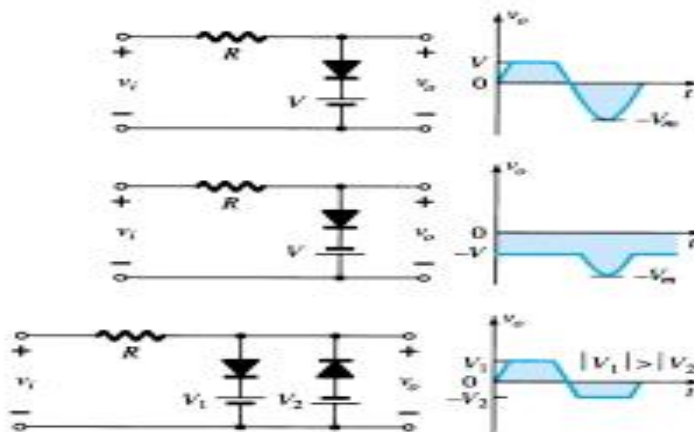
Biased Series Clippers (Ideal Diodes)



Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)



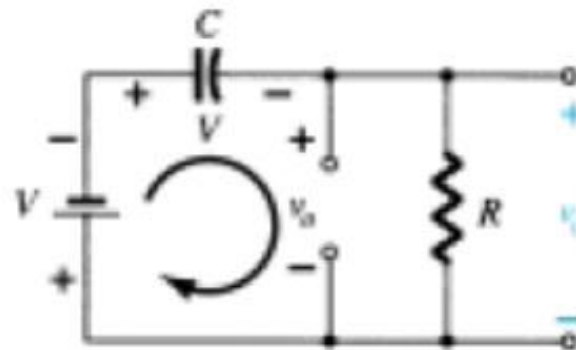
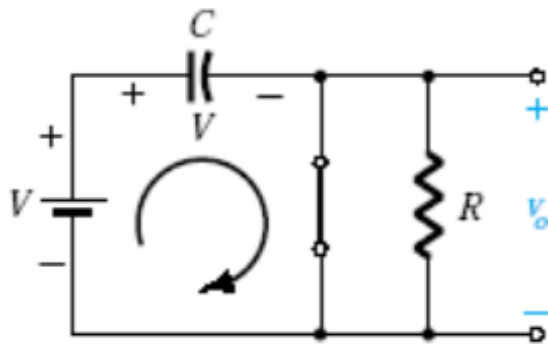
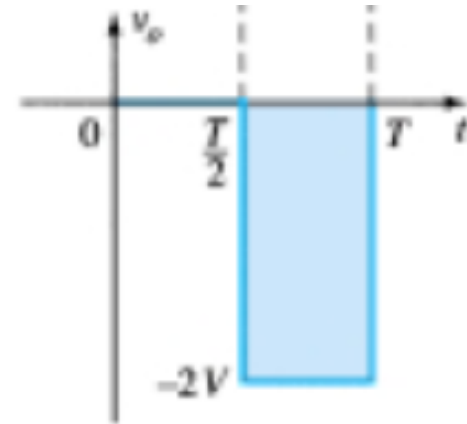
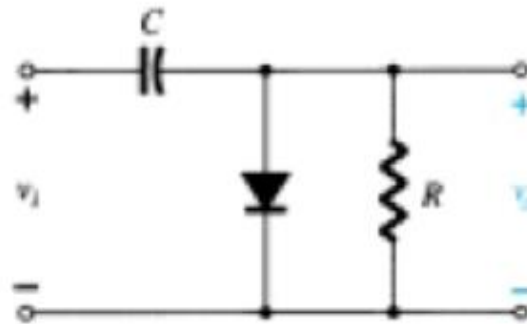
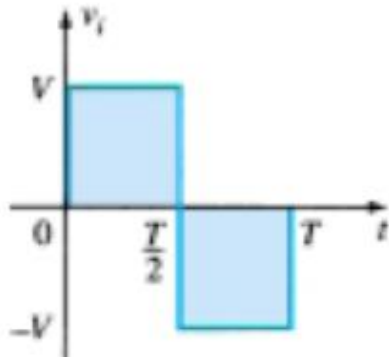
Clamper circuit

- Clamper adds a dc level to an ac signal. It shifts the ac signal to a certain dc level.
- Needs diode, resistor, capacitor.
- Sometimes it needs a dc battery too.

Steps to analyze

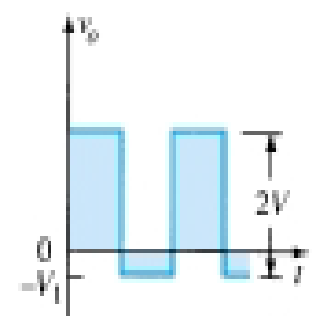
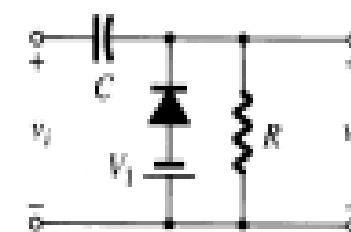
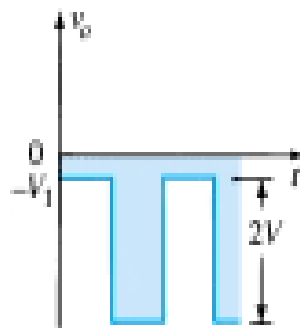
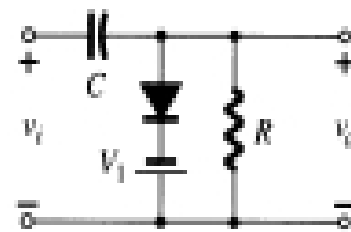
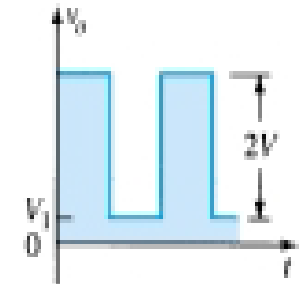
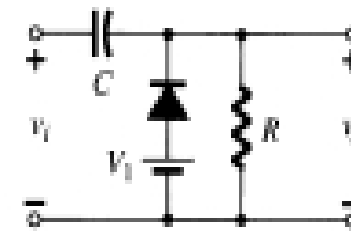
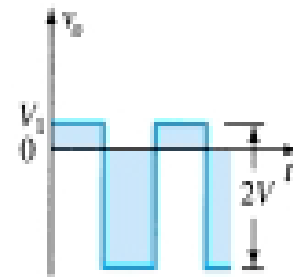
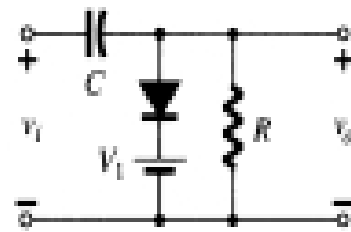
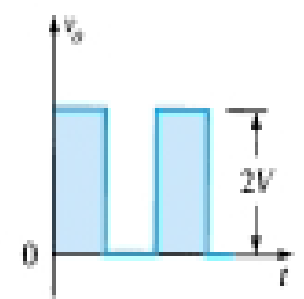
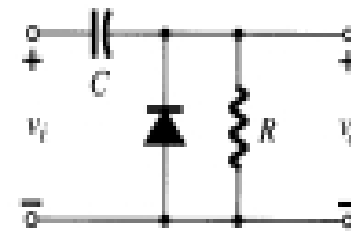
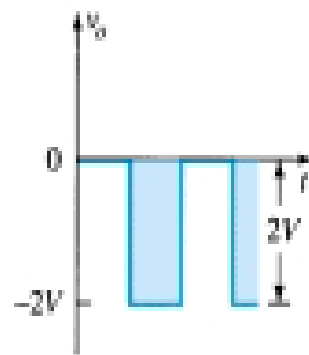
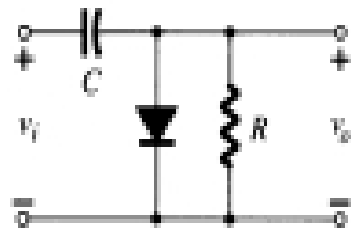
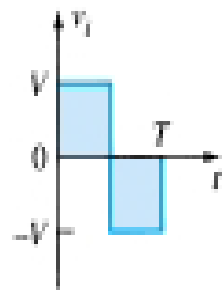
- Analyze the circuit for the portion of the input signal first for which the diode conducts.
- Find the voltage across the capacitor and indicate the polarity. Find the corresponding output voltage.
- Then, determine the output voltage due to the rest portion of the input signal.

Example-1



Applying KVL, $-V - V - v_o = 0$

$$v_o = -2V$$



Clamping network with a sinusoidal input

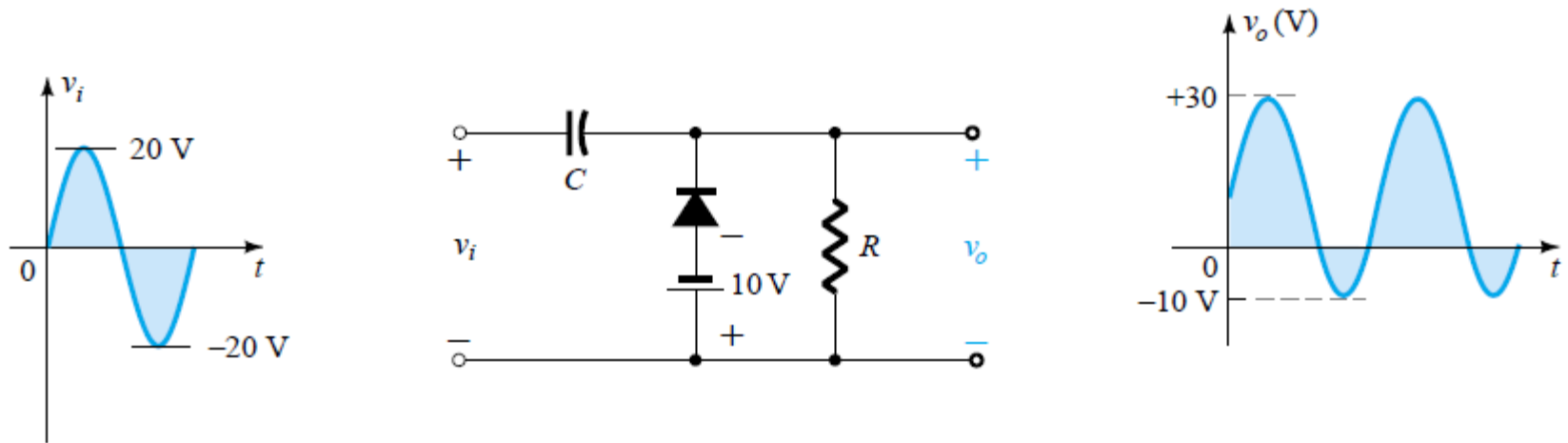
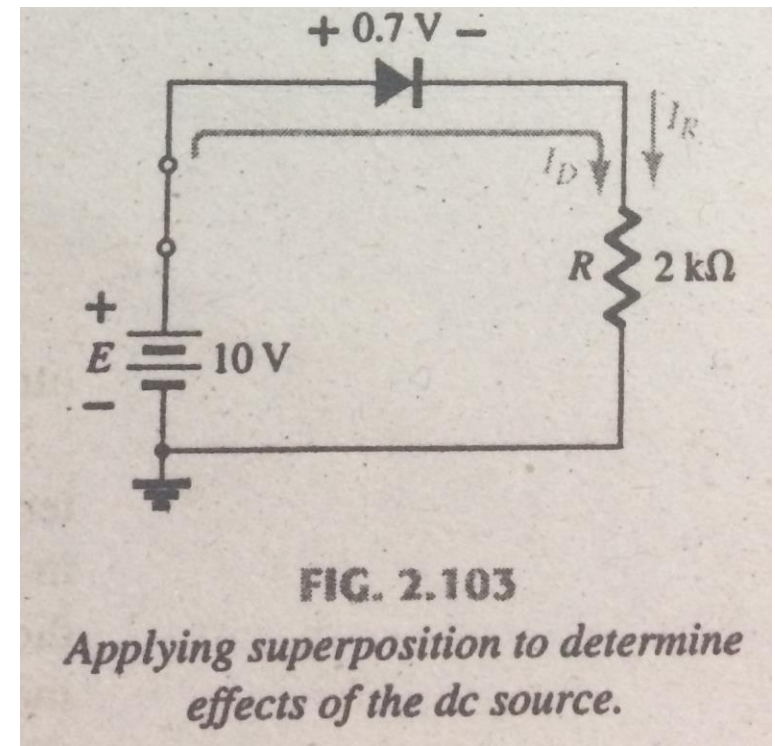
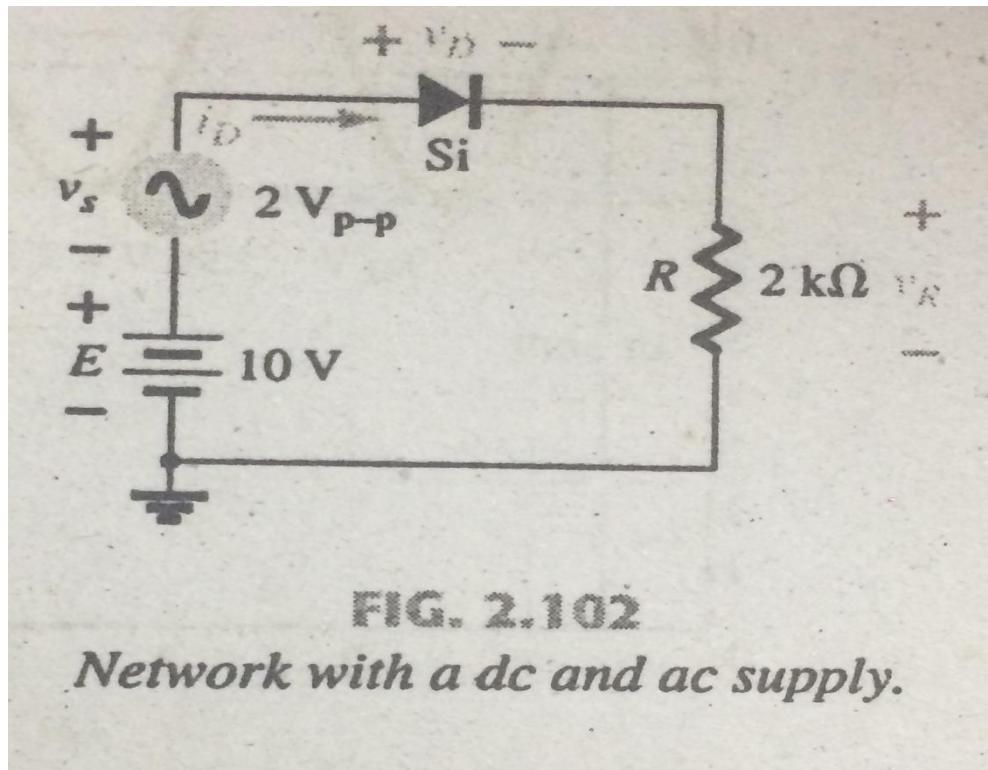


Figure 2.104 Clamping network with a sinusoidal input.

Network with a DC and AC source

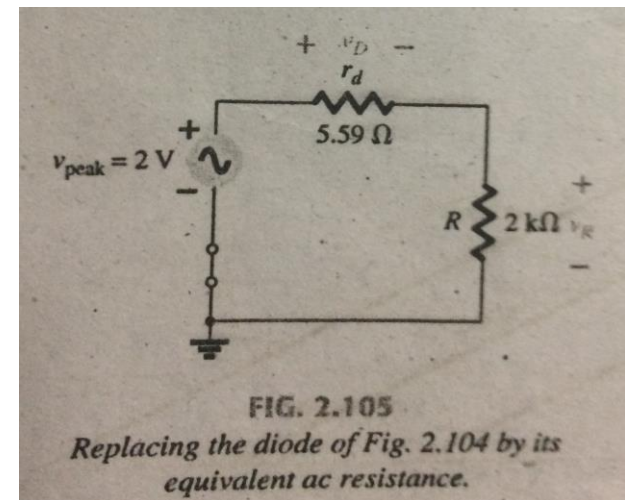
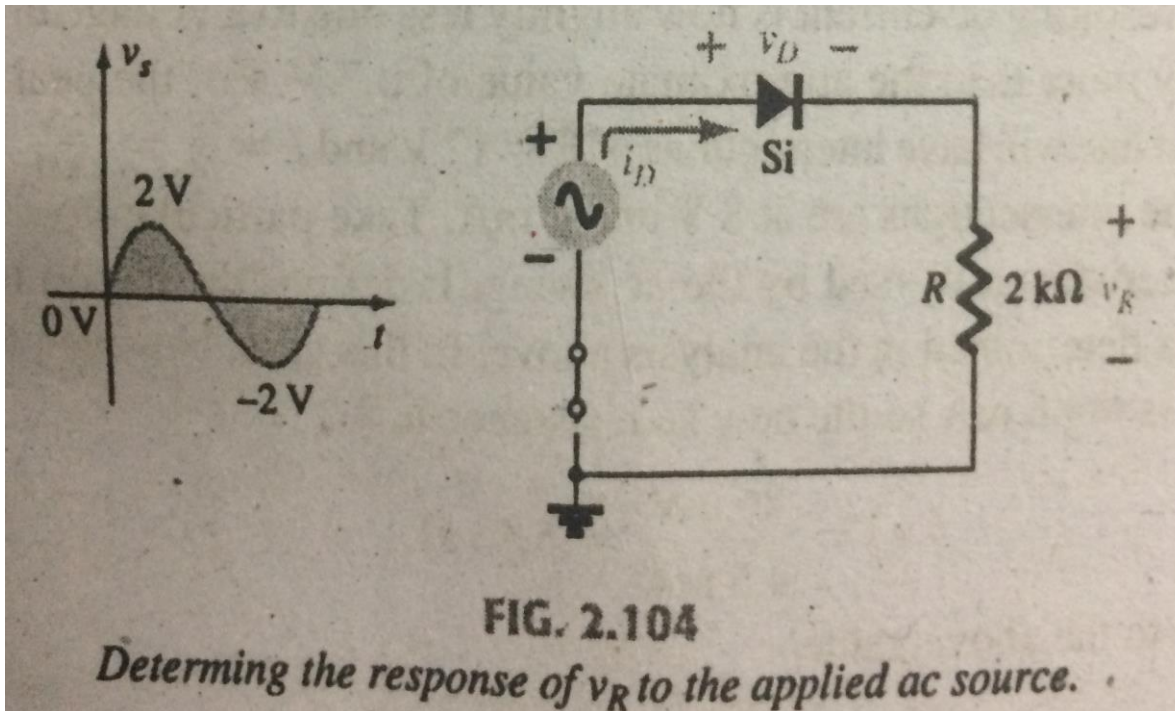
The response of any network with both an ac and a dc source can be found by finding the response to each source independently and then combining them.



$$V_R = E - V_D = 10\text{ V} - 0.7\text{ V} = 9.3\text{ V}$$
$$I_D = I_R = \frac{9.3\text{ V}}{2\text{ k}\Omega} = 4.65\text{ mA}$$

Network with a DC and AC source

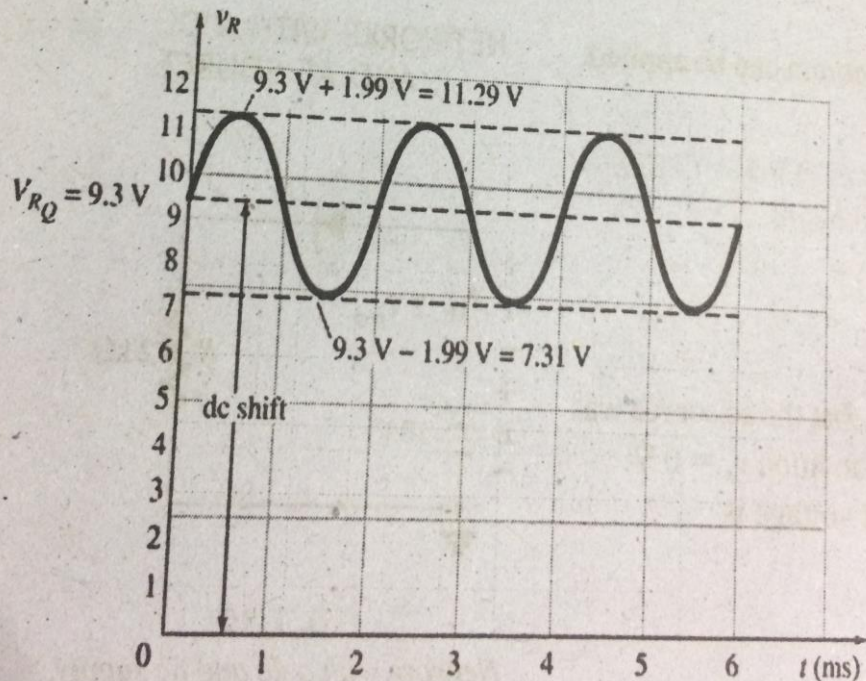
$$r_d = \frac{26 \text{ mV}}{I_D} = \frac{26 \text{ mV}}{4.65 \text{ mA}} = 5.59 \Omega$$



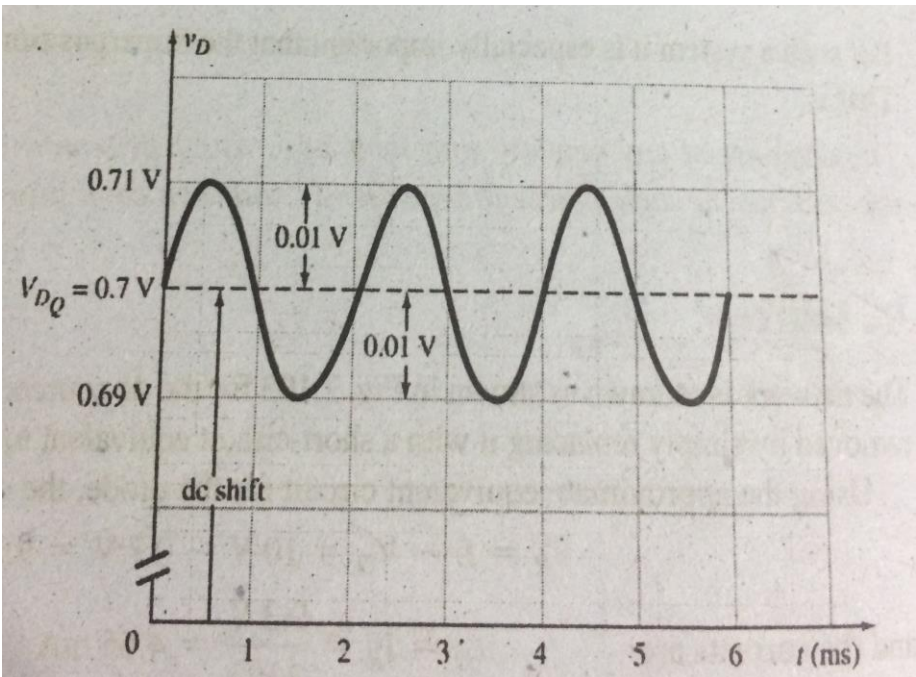
Network with a DC and AC source

$$V_{R_{peak}} = \frac{2 \text{ k}\Omega (2 \text{ V})}{2 \text{ k}\Omega + 5.59 \Omega} \cong 1.99 \text{ V}$$

$$V_{D_{peak}} = V_{S_{peak}} - V_{R_{peak}} = 2 \text{ V} - 1.99 \text{ V} = 0.01 \text{ V} = 10 \text{ mV}$$

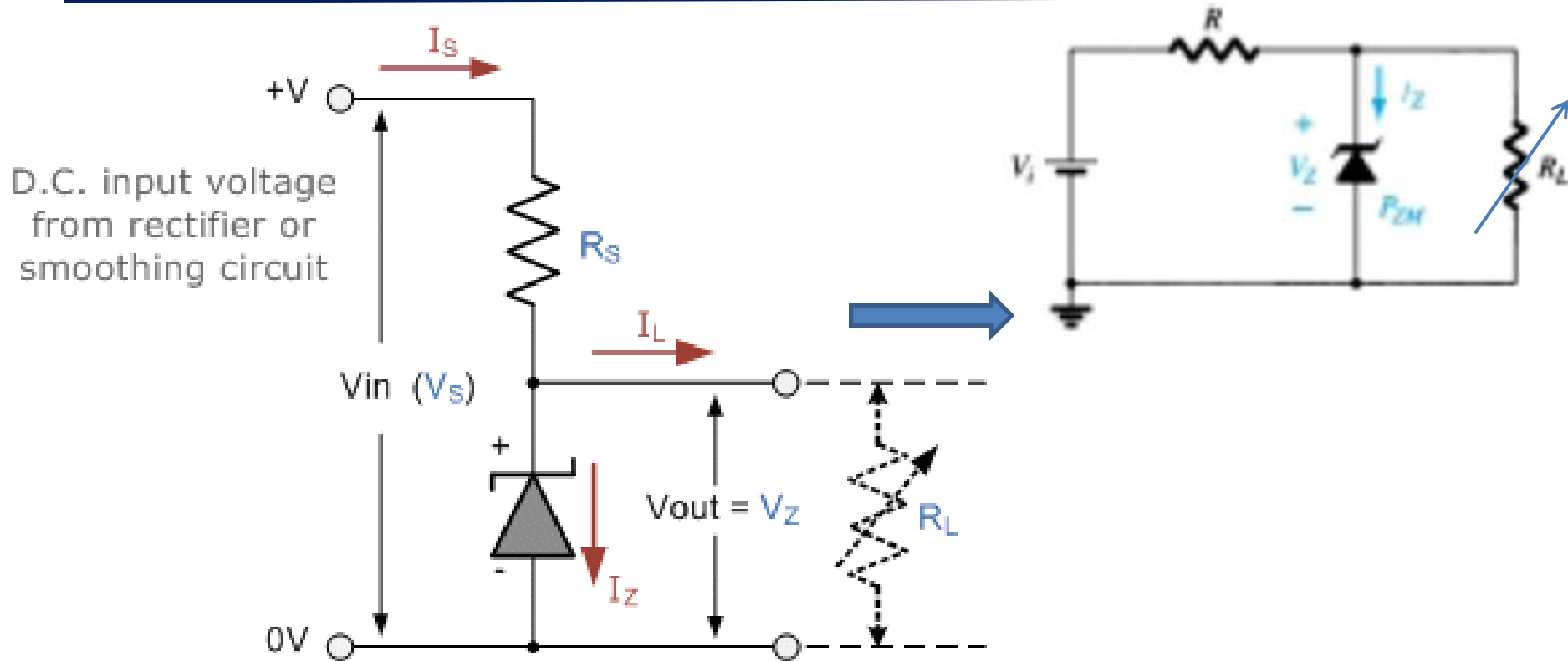


(a)



(b)

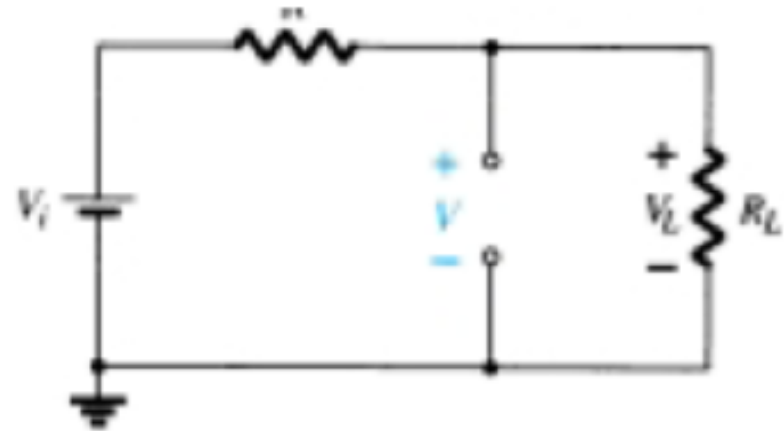
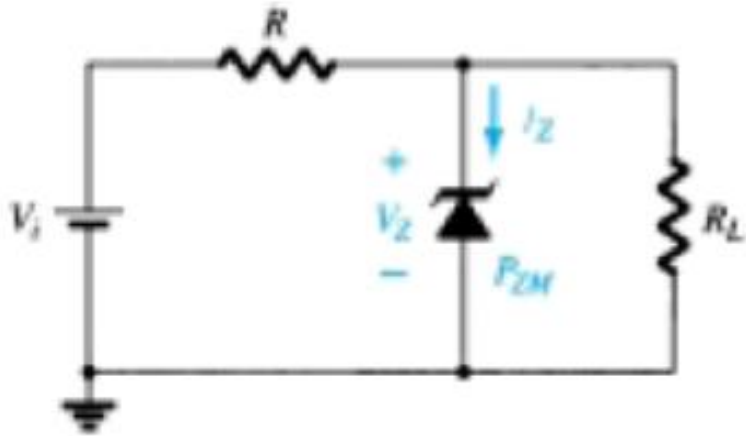
Regulator circuit



Three conditions are analysed for above Zener regulator circuit:

1. V_i and R fixed
2. Fixed V_i , variable R_L
3. Fixed R_L , Variable V_i

Regulator circuit (V_i and R fixed)



$$V = V_L = \frac{R_L V_i}{R + R_L}$$

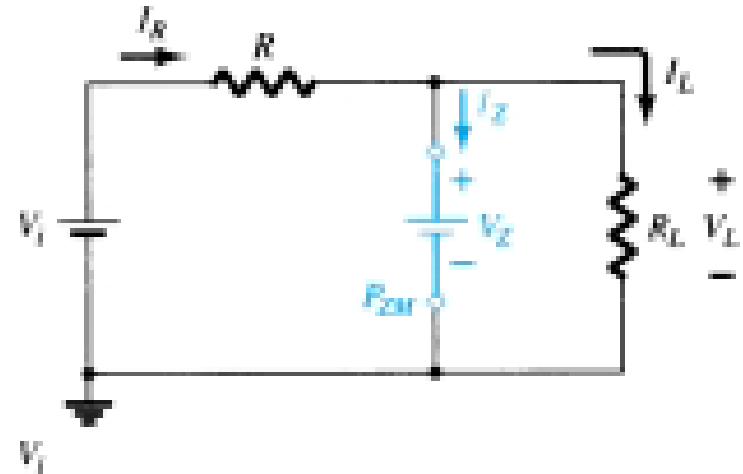
If $V \geq V_Z$, the zener is on and if less than V_Z then zener will be off.

Regulator circuit (V_i and R_L fixed)

$$V_L = V_Z$$

$$I_R = I_Z + I_L$$

$$I_Z = I_R - I_L$$



where

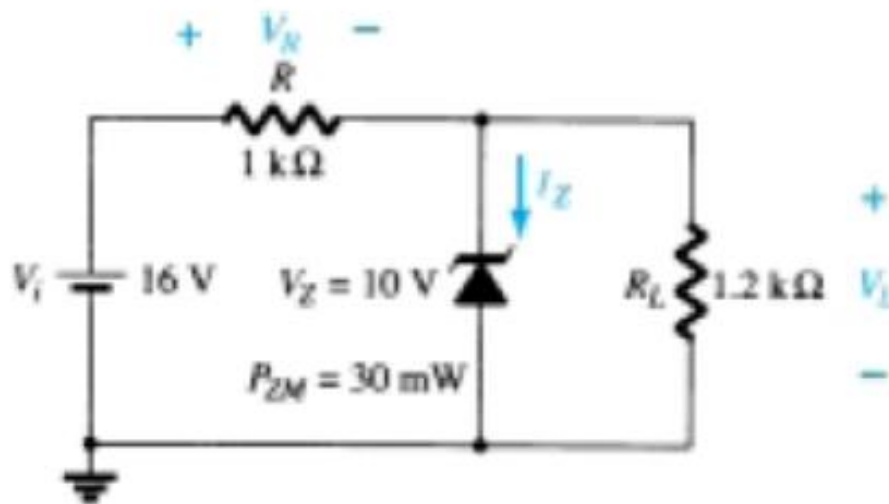
$$I_L = \frac{V_L}{R_L} \quad \text{and} \quad I_R = \frac{V_R}{R} = \frac{V_i - V_L}{R}$$

The power dissipated by the Zener diode is determined by

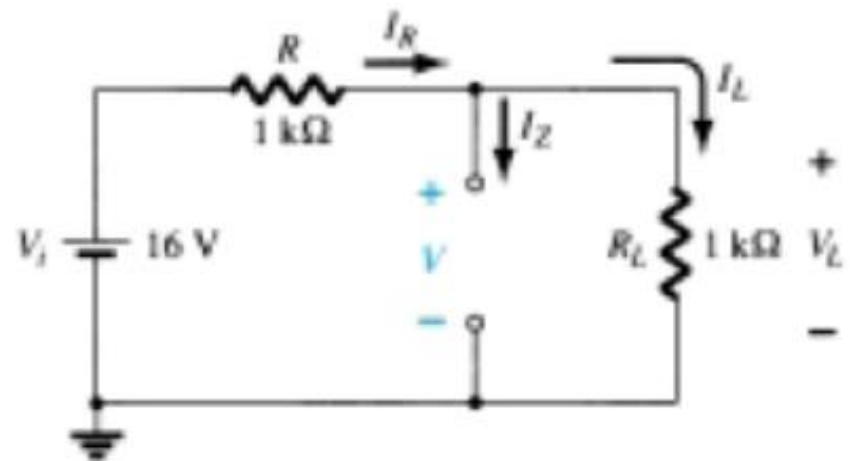
$$P_Z = V_Z I_Z$$

Example-1

- (a) For the Zener diode network of Fig. 2.109, determine V_L , V_R , I_Z , and P_Z .
 (b) Repeat part (a) with $R_L = 3 \text{ k}\Omega$.



$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$



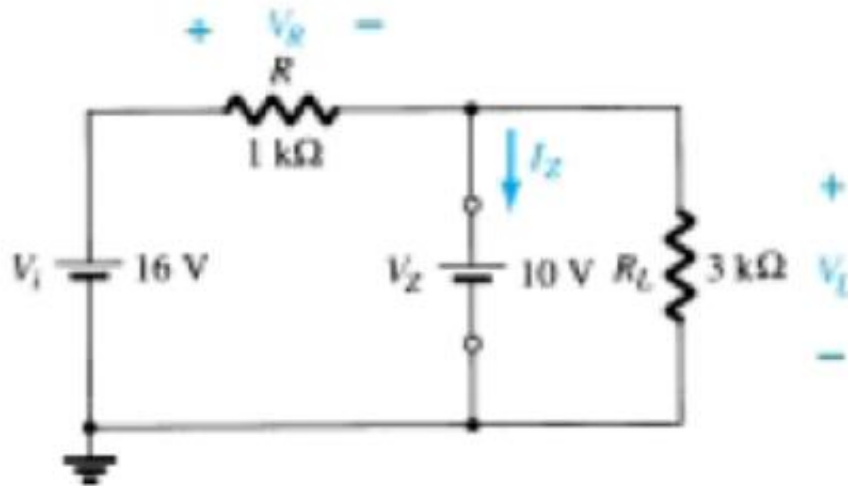
$$V_L = V = 8.73 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

$$I_Z = 0 \text{ A}$$

$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$

Solution to example-1



$$V = \frac{R_L V_1}{R + R_L} = \frac{3\text{ k}\Omega (16\text{ V})}{1\text{ k}\Omega + 3\text{ k}\Omega} = 12\text{ V}$$

$$V_L = V_Z = 10\text{ V}$$

$$V_R = V_1 - V_L = 16\text{ V} - 10\text{ V} = 6\text{ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{10\text{ V}}{3\text{ k}\Omega} = 3.33\text{ mA}$$

$$I_R = \frac{V_R}{R} = \frac{6\text{ V}}{1\text{ k}\Omega} = 6\text{ mA}$$

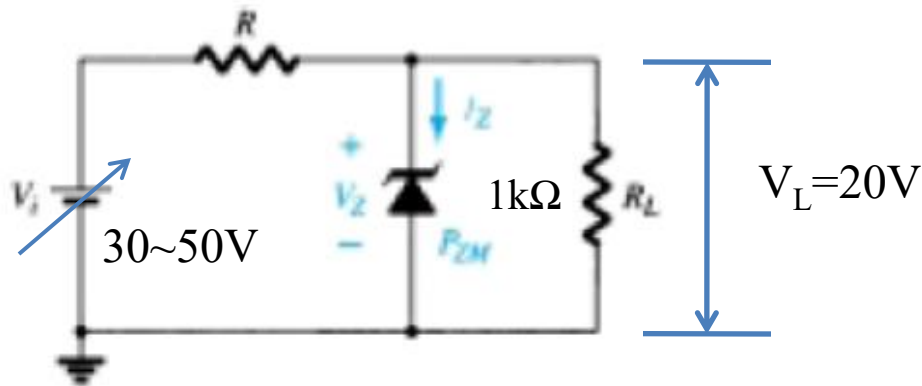
$$I_Z = I_R - I_L \text{ [Eq. (2.18)]}$$

$$= 6\text{ mA} - 3.33\text{ mA}$$

$$= 2.67\text{ mA}$$

$$P_Z = V_Z I_Z = (10\text{ V})(2.67\text{ mA}) = 26.7\text{ mW}$$

Example-2



At 30 V we have to be sure Zener diode is “on”.

$$\therefore V_L = 20 \text{ V} = \frac{R_L V_i}{R_L + R_s} = \frac{1 \text{ k}\Omega (30 \text{ V})}{1 \text{ k}\Omega + R_s}$$

Solving, $R_s = 0.5 \text{ k}\Omega$

$$\text{At } 50 \text{ V, } I_{R_s} = \frac{50 \text{ V} - 20 \text{ V}}{0.5 \text{ k}\Omega} = 60 \text{ mA, } I_L = \frac{20 \text{ V}}{1 \text{ k}\Omega} = 20 \text{ mA}$$

$$I_{ZM} = I_{R_s} - I_L = 60 \text{ mA} - 20 \text{ mA} = \mathbf{40 \text{ mA}}$$

Regulator circuit (Fixed V_i and Variable R_L)

$$V_L = V_Z = \frac{R_L V_i}{R_L + R}$$

$$R_{L_{\min}} = \frac{R V_Z}{V_i - V_Z}$$

$$I_{L_{\max}} = \frac{V_L}{R_L} = \frac{V_Z}{R_{L_{\min}}}$$

Once the diode is in the “on” state, the voltage across R remains fixed at

$$V_R = V_i - V_Z$$

and I_R remains fixed at

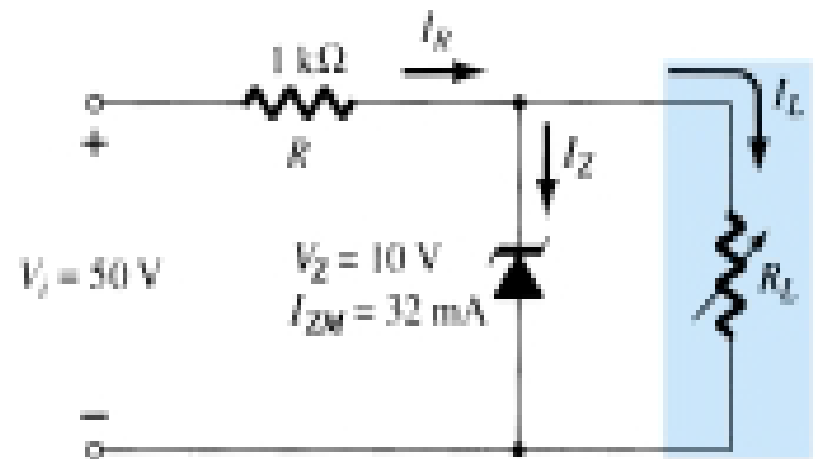
$$I_R = \frac{V_R}{R}$$

The Zener current

$$I_Z = I_R - I_L$$

$$I_{L_{\min}} = I_R - I_{ZM}$$

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}}$$



Regulator circuit (Fixed R_L , Variable V_i)

EXAMPLE 2.28

Determine the range of values of V_i that will maintain the Zener diode of Fig. 2.115 in the “on” state.

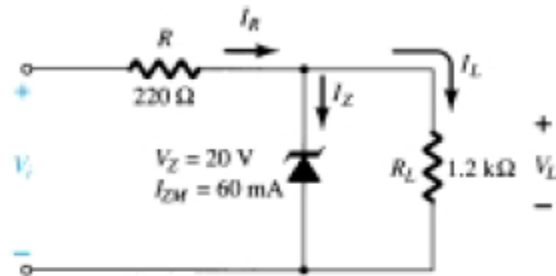


Figure 2.115 Regulator for Example 2.28.

Solution

$$\text{Eq. (2.27): } V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200 \, \Omega + 220 \, \Omega)(20 \, \text{V})}{1200 \, \Omega} = 23.67 \, \text{V}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20 \, \text{V}}{1.2 \, \text{k}\Omega} = 16.67 \, \text{mA}$$

$$\begin{aligned} \text{Eq. (2.28): } I_{R_{\max}} &= I_{ZM} + I_L = 60 \, \text{mA} + 16.67 \, \text{mA} \\ &= 76.67 \, \text{mA} \end{aligned}$$

$$\begin{aligned} \text{Eq. (2.29): } V_{i_{\max}} &= I_{R_{\max}} R + V_Z \\ &= (76.67 \, \text{mA})(0.22 \, \text{k}\Omega) + 20 \, \text{V} \\ &= 16.87 \, \text{V} + 20 \, \text{V} \\ &= 36.87 \, \text{V} \end{aligned}$$

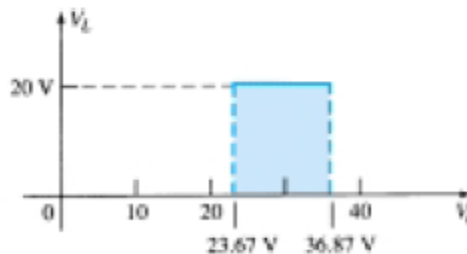


Figure 2.116 V_L versus V_i for the regulator of Fig. 2.115

A plot of V_L versus V_i is provided in Fig. 2.116.

Half-wave Voltage Doubler

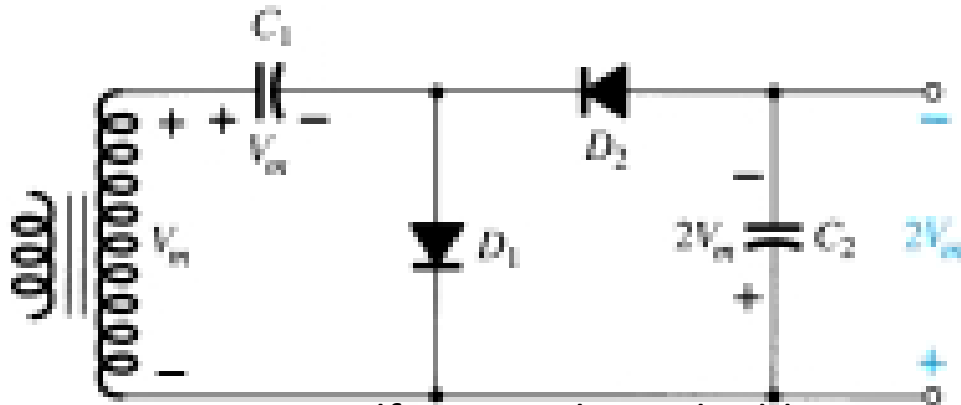


Figure 2.121 Half-wave voltage doubler.

$$-V_m - V_{C_1} + V_{C_2} = 0$$

$$-V_m - V_m + V_{C_2} = 0$$

from which

$$V_{C_2} = 2V_m$$

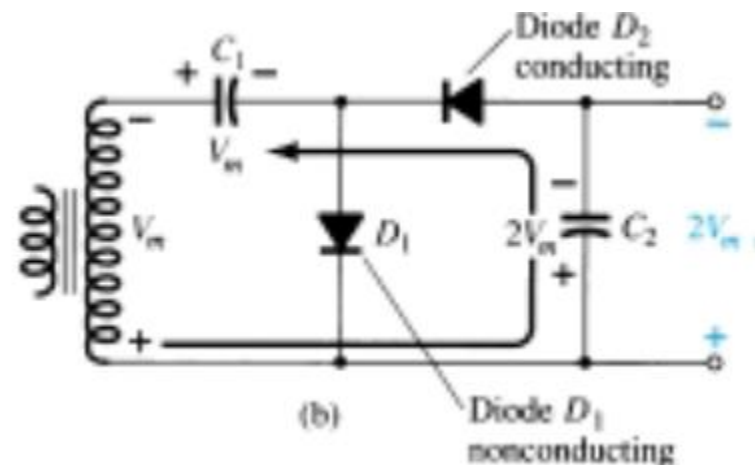
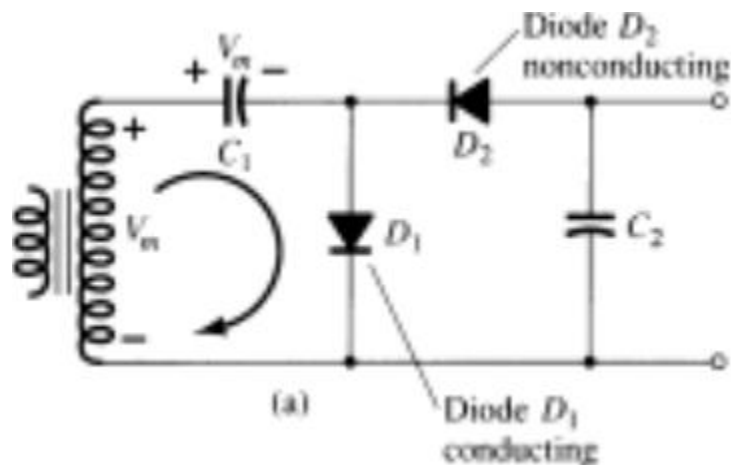


Figure 2.122 Double operation, showing each half-cycle of operation: (a) positive half-cycle; (b) negative half cycle.

Voltage Tripler and Quadrupler

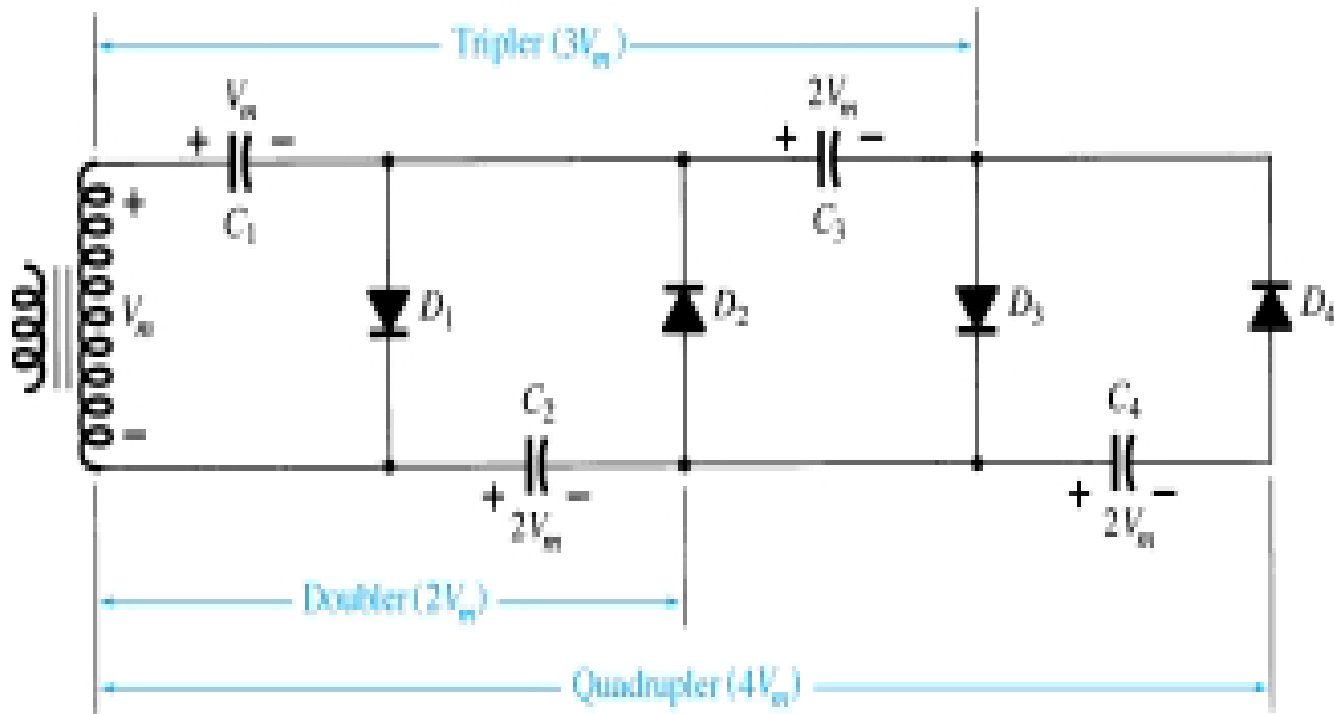
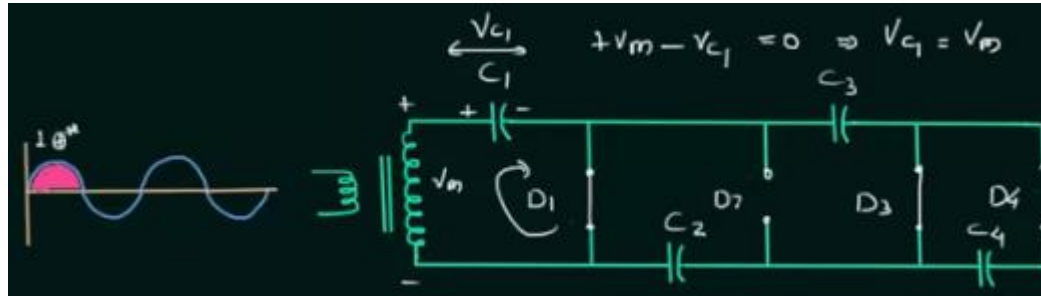


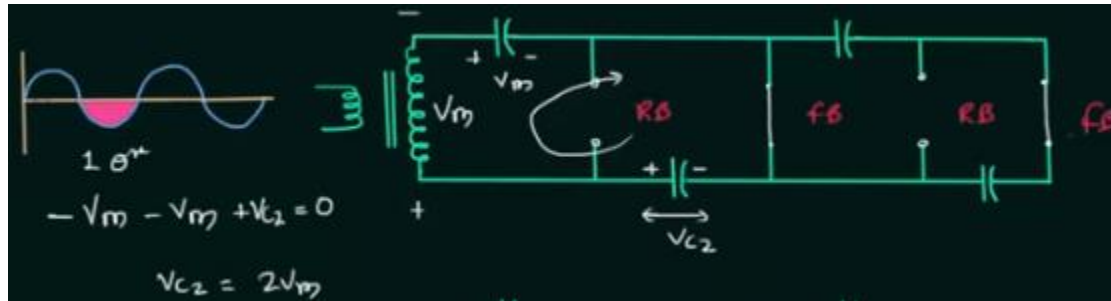
Figure 2.125 Voltage tripler and quadrupler.

Voltage Tripler and Quadrupler

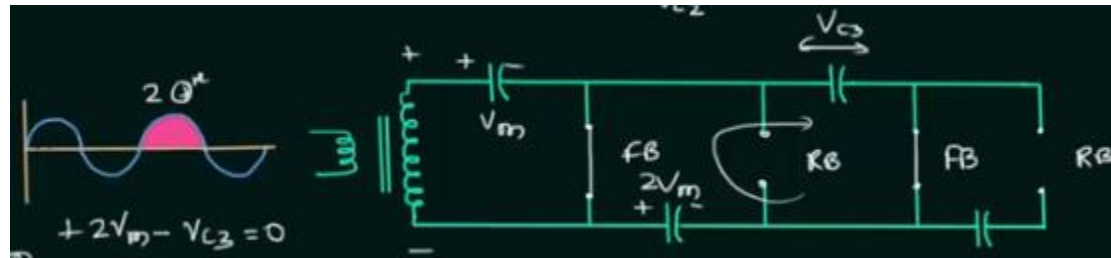
Step: 01



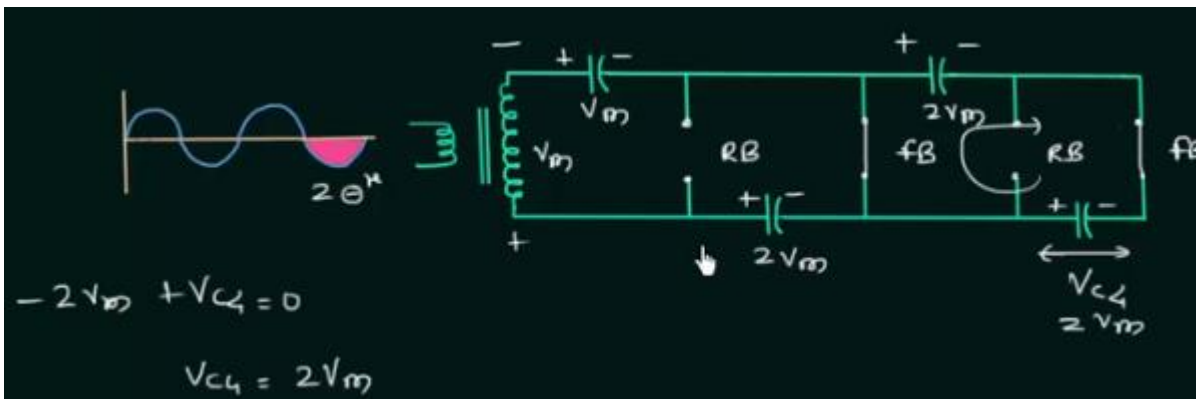
Step: 02



Step: 03



Step: 04



Practical Applications Summary

- Rectification
- Protective configurations
- Polarity insurance
- Controlled battery-powered backup
- Polarity detector
- Ac regulator and square-wave generator

Thank You